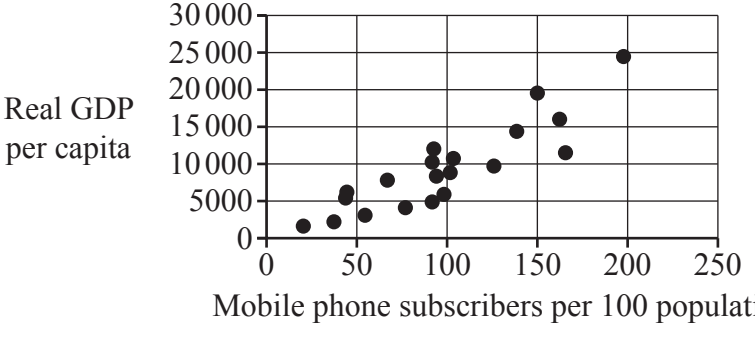
5 The pre-release material contains information for countries in the world concerning real GDP per capita in US$ and mobile phone subscribers per 100 population. In an investigation into the relationship between these two variables, a student takes a sample of 20 countries in Africa. The student draws a scatter diagram for the data, which is shown in **Fig. 5.1**.

Fig. 5.1

Africa 1st sample

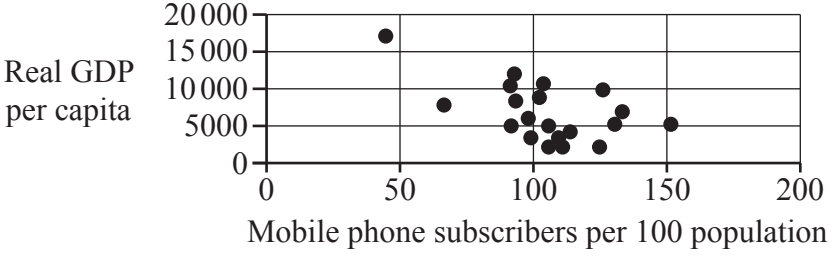


(a) What does **Fig. 5.1** suggest about the relationship between real GDP per capita and the number of mobile phone subscribers per 100 population? **[1]**

Another student collects a **different** sample of 20 countries from Africa, and draws a scatter diagram for the data, which is shown in **Fig. 5.2**.

Fig. 5.2

Africa 2nd sample



(b) What does **Fig. 5.2** suggest about the relationship between real GDP per capita and the number of mobile phone subscribers per 100 population? **[1]**

(c) Explain whether either of the two scatter diagrams is likely to be representative of the true relationship between real GDP per capita and the number of mobile phone subscribers per 100 population, for countries in Africa. **[1]**